

Gle1 is required for tRNA to stimulate Dbp5 ATPase activity *in vitro* and to promote Dbp5 mediated tRNA export *in vivo*

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Abstract

Cells must maintain a pool of processed and charged transfer RNAs (tRNA) to sustain translation capacity and efficiency. Numerous parallel pathways support the processing and directional movement of tRNA in and out of the nucleus to meet this cellular demand. Recently, several proteins known to control messenger RNA (mRNA) transport were implicated in tRNA export. The DEAD-box Protein 5, Dbp5, is one such example. In this study, genetic and molecular evidence demonstrates that Dbp5 functions parallel to the canonical tRNA export factor Los1. *In vivo* co-immunoprecipitation data further shows Dbp5 is recruited to tRNA independent of Los1, Msn5 (another tRNA export factor), or Mex67 (mRNA export adaptor), which contrasts with Dbp5 recruitment to mRNA that is abolished upon loss of Mex67 function. However, as with mRNA export, overexpression of Dbp5 dominant-negative mutants indicates a functional ATPase cycle and that binding of Dbp5 to Gle1 is required by Dbp5 to direct tRNA export. Biochemical characterization of the Dbp5 catalytic cycle demonstrates the direct interaction of Dbp5 with tRNA (or double stranded RNA) does not activate Dbp5 ATPase activity, rather tRNA acts synergistically with Gle1 to fully activate Dbp5. These data suggest a model where Dbp5 directly binds tRNA to mediate export, which is spatially regulated via Dbp5 ATPase activation at nuclear pore complexes by Gle1.

eLife assessment

The work is a **useful** contribution to understanding the mechanism of nuclear export of tRNA in budding yeast. The authors report that Dbp5 functions in parallel with Los1 in tRNA export, in a manner dependent on Gle1 and requiring the ATPase cycle of Dbp5, but independent of Mex67, Dbp5's partner in mRNA export. The evidence for this conclusion is still **incomplete**, as is the biochemical evidence that Dbp5 interacts directly with tRNA *in vitro* with Gle1 and co-factor InsP6 triggering Dbp5 ATPase activity in the Dbp5-tRNA complex. The evidence that Dbp5 interacts with tRNA in cells independently of Los1, Msn5 and Mex67 is, however, **solid**.

Key to the production of proteins is the delivery of amino acids to ribosomes by transfer RNAs (tRNAs). Precursor tRNAs (pre-tRNAs) are first transcribed by RNA Polymerase III with a 5' leader and a 3' trailer sequence (Guerrier-Takada, Gardiner et al. 1983 [↗](#), Cook, Fukuhara et al. 2009 [↗](#), Skowronek, Grzechnik et al. 2014 [↗](#), Harris 2016 [↗](#), Graczyk, Ciesla et al. 2018 [↗](#)). To function in translation, pre-tRNAs must be end matured and modified, spliced, amino-acylated and exported from the nucleus in eukaryotes. Importantly, it has come to be appreciated that these steps in tRNA processing are not necessarily sequential (Trotta, Miao et al. 1997 [↗](#), Yoshihisa, Yunoki-Esaki et al. 2003 [↗](#), Yoshihisa, Ohshima et al. 2007 [↗](#), Wu and Hopper 2014 [↗](#), Hopper and Huang 2015 [↗](#), Wu, Bao et al. 2015, Phizicky and Hopper 2023 [↗](#)). For example, in the budding yeast *Saccharomyces cerevisiae*, intron-containing tRNAs are spliced in the cytoplasm on the mitochondrial surface by the tRNA Splicing Endonuclease (SEN) complex following nuclear export (Yoshihisa, Yunoki-Esaki et al. 2003 [↗](#), Yoshihisa, Ohshima et al. 2007 [↗](#), Skowronek, Grzechnik et al. 2014, Wan and Hopper 2018 [↗](#)). Furthermore, cytoplasmic tRNAs can be reimported to the nucleus through retrograde transport to allow tRNAs to undergo further modification and quality control which is also used to repress translation during conditions of stress (Murthi, Shaheen et al. 2010, Hopper and Huang 2015 [↗](#), Nostramo and Hopper 2020 [↗](#)). As tRNAs cycle between the nucleus and cytoplasm, they undergo various chemical modifications that are also spatially regulated (Phizicky and Hopper 2023 [↗](#)). The proper processing and modification of tRNAs through this complex regulatory network dictates proper folding of pre-tRNAs and recognition by appropriate tRNA binding proteins that control subsequent steps in the tRNA life cycle.

In the context of nuclear export, end maturation and amino-acylation are required for recognition by the tRNA Exportin Los1/Exportin-t and Msn5 respectively (Huang and Hopper 2015 [↗](#), Chatterjee, Majumder et al. 2017, Nostramo and Hopper 2020 [↗](#), Chatterjee, Marshall et al. 2022). Although Los1 and Msn5 are the best characterized tRNA export factors, both are non-essential and can be deleted in combination (Takano, Endo et al. 2005, Murthi, Shaheen et al. 2010). Given the essential nature of tRNA export, this has prompted several studies, including a comprehensive screen of nearly all annotated genes in *S. cerevisiae*, to identify additional factors responsible for regulating nucleocytoplasmic dynamics of tRNA localization and processing (Feng and Hopper 2002 [↗](#), Steiner-Mosonyi, Leslie et al. 2003, Eswara, McGuire et al. 2009, Huang and Hopper 2015 [↗](#), Wu, Bao et al. 2015, Chatterjee, Majumder et al. 2017, Nostramo and Hopper 2020 [↗](#), Chatterjee, Marshall et al. 2022). These studies have elucidated functional roles of the karyopherin Crm1 and mobile nucleoporin Mex67 in regulating tRNA nucleocytoplasmic dynamics (Chatterjee, Majumder et al. 2017, Derrer, Mancini et al. 2019, Nostramo and Hopper 2020 [↗](#), Chatterjee, Marshall et al. 2022). Both factors have previously been shown to function in messenger RNA (mRNA) export (Hodge, Colot et al. 1999, Takemura, Inoue et al. 2004, Lund and Guthrie 2005 [↗](#), Derrer, Mancini et al. 2019), with Mex67 serving an essential role in the process (Hodge, Colot et al. 1999, Lund and Guthrie 2005 [↗](#), Adams and Wente 2020 [↗](#)). In tRNA export, it is thought that Crm1 and Mex67 both function parallel to Los1 to support export of pre-tRNA and re-export of spliced tRNA that have undergone retrograde transport (Chatterjee, Majumder et al. 2017, Nostramo and Hopper 2020 [↗](#), Chatterjee, Marshall et al. 2022). Moreover, recent reports suggest subspecies specialization and family preferences for tRNA cargo, with Mex67 serving a unique function in the premature export of pre-tRNAs that have not completed 5' end processing (Chatterjee, Marshall et al. 2022).

While non-coding RNA (ncRNA) and mRNA export pathways have long been hypothesized to be regulated by independent processes, recent studies indicate the participation of numerous mRNA export factors in regulation of diverse ncRNAs (Yao, Roser et al. 2007, Yao, Lutzmann et al. 2008, Faza, Chang et al. 2012, Bai, Moore et al. 2013, Wu, Becker et al. 2014, Becker, Hirsch et al. 2019, Vasiyanovich, Bajon et al. 2020). In addition to tRNA and mRNA, both Mex67 and Crm1 have functions in pre-snRNA, pre-ribosomal subunit, and *TLC1* (telomerase) export (Yao, Roser et al. 2007, Faza, Chang et al. 2012, Bai, Moore et al. 2013, Wu, Becker et al. 2014, Becker, Hirsch et al.

2019, Vasianovich, Bajon et al. 2020). Similarly, in addition to Mex67 and Crm1, the DEAD-box Protein 5 (Dbp5) has also been implicated in each of these RNA export pathways, including tRNA export (Wu, Becker et al. 2014, Neumann, Wu et al. 2016, Mikhailova, Shuvalova et al. 2017, Becker, Hirsch et al. 2019, Lari, Arul Nambi Rajan et al. 2019). Dbp5 is most well known as an RNA-stimulated ATPase that is spatially regulated by the nucleoporins Nup159 and Gle1 with the small molecule inositol hexakisphosphate (InsP₆) at the cytoplasmic face of the nuclear pore complex (NPC) to promote the directional export of mRNA (Snay-Hodge, Colot et al. 1998, Tseng, Weaver et al. 1998, Hodge, Colot et al. 1999, Schmitt, von Kobbe et al. 1999, Strahm, Fahrenkrog et al. 1999, Zhao, Jin et al. 2002, Weirich, Erzberger et al. 2004, Lund and Guthrie 2005 [\[1\]](#), Alcazar-Roman, Tran et al. 2006 [\[2\]](#), Weirich, Erzberger et al. 2006, Tran, Zhou et al. 2007, Dossani, Weirich et al. 2009, Fan, Cheng et al. 2009, von Moeller, Basquin et al. 2009, Hodge, Tran et al. 2011, Montpetit, Thomsen et al. 2011, Noble, Tran et al. 2011, Kaminski, Siebrasse et al. 2013, Wong, Cao et al. 2016, Adams, Mason et al. 2017 [\[3\]](#), Wong, Gray et al. 2018, Adams and Wente 2020 [\[4\]](#)). In addition to these export roles, Dbp5 also shuttles between the nucleus and cytoplasm with reported roles in transcription, R-loop metabolism, and translation (Hodge, Colot et al. 1999, Estruch and Cole 2003 [\[5\]](#), Gross, Siepmann et al. 2007, Scarcelli, Viggiano et al. 2008, Tieg and Krebber 2013 [\[6\]](#), Mikhailova, Shuvalova et al. 2017, Lari, Arul Nambi Rajan et al. 2019, Beissel, Grosse et al. 2020). Notably, Nup159 and Gle1 mutants also alter tRNA export (Lari, Arul Nambi Rajan et al. 2019), suggesting the mRNA export pathway as a whole (e.g., Mex67, Dbp5, Gle1/ InsP₆, Nup159) may support tRNA export. However, a mechanistic understanding of how Dbp5 supports these diverse functions and whether they are regulated by similar co-factors and enzymatic activity is largely not understood.

A recent mutagenesis screen within our research group identified mutants that alter the nucleocytoplasmic shuttling dynamics of Dbp5 (Lari, Arul Nambi Rajan et al. 2019). Mutations were identified (e.g., *dbp5*^{L12A}) that disrupt a nuclear export sequence (NES) recognized by Crm1 to promote Dbp5 transport out of the nucleus. In contrast, another mutant, *dbp5*^{R423A}, was found to be impaired in nuclear import from the cytoplasm. Importantly, neither of these mutations disrupt the essential mRNA export functions of Dbp5; however, limited nuclear access of *dbp5*^{R423A} induced tRNA export defects suggesting a potentially novel role for nuclear Dbp5 in tRNA export. In support of this hypothesis, a physical interaction between Dbp5 and tRNA was also characterized, which was elevated upon nuclear accumulation of Dbp5 in a *dbp5*^{L12A} mutant (Lari, Arul Nambi Rajan et al. 2019).

In this study, genetic and biochemical characterization of Dbp5 mediated tRNA export was performed. The data show that Dbp5 is recruited to pre-tRNA independent of Los1, Msn5, and Mex67, positioning the DEAD-box protein function parallel to the primary Los1 mediated pre-tRNA export pathway. In contrast to single stranded RNA substrates (e.g., mRNA), this study further demonstrates an interaction of Dbp5 with tRNA and double stranded RNA (dsRNA) that leads to robust ATPase activation only in the presence of Gle1/ InsP₆. These findings, together with previous research (Lari, Arul Nambi Rajan et al. 2019), suggest that Dbp5 engages tRNA within the nucleus in a manner distinct from mRNA, and requires Gle1/ InsP₆ to mediate RNA stimulated ATPase activation of Dbp5 to promote export.

Results

Dbp5 functions parallel to Los1 in pre-tRNA export

Previous studies have placed the mRNA export factor Mex67, a proposed target of Dbp5 regulation in mRNA export (Lund and Guthrie 2005 [\[1\]](#), Adams and Wente 2020 [\[4\]](#)), in a pathway parallel to Los1 regulated pre-tRNA nucleo-cytoplasmic shuttling (Chatterjee, Majumder et al. 2017). To investigate the relationship between Dbp5 and Los1, it was first determined if the subcellular distribution of either protein was dependent on the activity of the other. In a *los1Δ*, the NPC

localization of GFP-Dbp5 was not altered (**Figure 1A** [↗](#)). Similarly, when a Dbp5 loss of function was induced using Auxin Induced Degradation (AID) (Nishimura, Fukagawa et al. 2009), Los1-GFP localization remained unchanged (**Figure 1B** [↗](#)) while mRNA export defects were detected (**Figure 1C** [↗](#)), confirming depletion of Dbp5 to a level sufficient to disrupt mRNA export. To test the genetic and phenotypic relationship of Los1 and Dbp5, a *los1Δ* was combined with the previously identified Dbp5 shuttling mutants (i.e., *dbp5^{L12A}* and *dbp5^{R423}*). Like a *mex67-5/los1Δ* strain (Chatterjee, Majumder et al. 2017), double mutants of *los1Δ* with *dbp5^{L12A}* (increased in the nucleus) or *dbp5^{R423}* (depleted from the nucleus) exhibited no growth defects (**Figure 1D** [↗](#)). In contrast, by tRNA FISH using a probe specific to the tRNA^{Ile}_{UAU} intron (SRIM03), the *los1Δ dbp5^{R423A}* double mutant exhibited significantly stronger nuclear accumulation of pre-tRNA as compared to single mutants or control after a 4-hour incubation at 37°C (**Figure 1E** [↗](#)/F). Notably, elevating nuclear pools of Dbp5 by combining *dbp5^{L12A}* with *los1Δ* did not alter the pre-tRNA export defects observed in *los1Δ* strains, indicating that excess nuclear Dbp5 is not sufficient to suppress the *los1Δ* phenotype. To complement the FISH data, Northern blotting analyses were conducted to assess tRNA processing in the double mutants. Intron-containing pre-tRNAs are transcribed with 5' leader and 3' trailer sequences to generate precursor molecules (P) that are end-processed in the nucleus to generate intron-containing intermediates (I) that can be detected by Northern blotting (Wu, Bao et al. 2015). When nuclear export of the end-processed pre-tRNA is disrupted a change in the precursor (P) to intron-containing intermediate (I) ratio is observed (Wu, Huang et al. 2013, Wu, Bao et al. 2015, Chatterjee, Majumder et al. 2017). As such, I/P ratios were measured in the *los1Δ/dbp5* double mutants. In line with the increased nuclear localization of pre-tRNAs observed by FISH, an additive defect in pre-tRNA processing was observed when *los1Δ* was combined *dbp5^{R423A}* (~2-fold increase in the I/P ratio relative to *los1Δ* alone, **Figure 1G** [↗](#)/H). These FISH and Northern blotting data mirror phenotypes previously reported for mutants of Mex67 in relation to Los1 (Chatterjee, Marshall et al. 2022), and ultimately place Dbp5 in a tRNA export pathway that has functions parallel to Los1.

Known tRNA export factors do not recruit Dbp5 to pre-tRNAs in vivo

Dbp5, like other DEAD-box proteins, has been reported to bind nucleic acids through sequence independent interactions with the phosphate backbone (Andersen, Ballut et al. 2006, Bono, Ebert et al. 2006, Sengoku, Nureki et al. 2006, Montpetit, Thomsen et al. 2011). For other helicases of the SF2 family, specificity in RNA substrates is conferred by adaptor proteins (Lund and Guthrie 2005 [↗](#), Jankowsky 2011 [↗](#), Thoms, Thomson et al. 2015). For this reason, it was investigated whether known tRNA export factors mediate recruitment of Dbp5 to pre-tRNA substrates. To do so, RNA immunoprecipitation (RIP) experiments were performed with protein-A (prA) tagged Dbp5, integrated at its endogenous locus and present as the sole copy of the gene, in strains where tRNA export factors were deleted. Co-immunoprecipitated RNAs were analyzed by RT-qPCR with primers specific to un-spliced intron-containing pre-tRNA^{Ile}_{UAU}. The abundance of the pre-tRNA^{Ile}_{UAU} target in each IP was normalized to the abundance of the target in the corresponding input sample to control for changes in gene expression. Relative enrichment of the target RNA was then compared to the background signal obtained from RNA IPs in a common untagged control. In a *los1Δ* strain, a ~2-fold reduction in the relative amount of pre-tRNA co-immunoprecipitated with Dbp5 (~9.5-fold enrichment in WT to ~4.5-fold in *los1Δ*) was observed, but the deletion failed to abolish the Dbp5 pre-tRNA *in vivo* interaction (**Figure 2A** [↗](#)). Loss of Msn5, which does not have a published role in the export of intron-containing pre-tRNAs (Huang and Hopper 2015 [↗](#)), did not cause a significant change in Dbp5 pre-tRNA interactions by RNA IP. These results are consistent with the additive tRNA export defects observed in *los1Δ dbp5^{R423A}* double mutants reported in **Figure 1** [↗](#) and support the hypothesis that Dbp5 functions in tRNA export parallel to Los1.

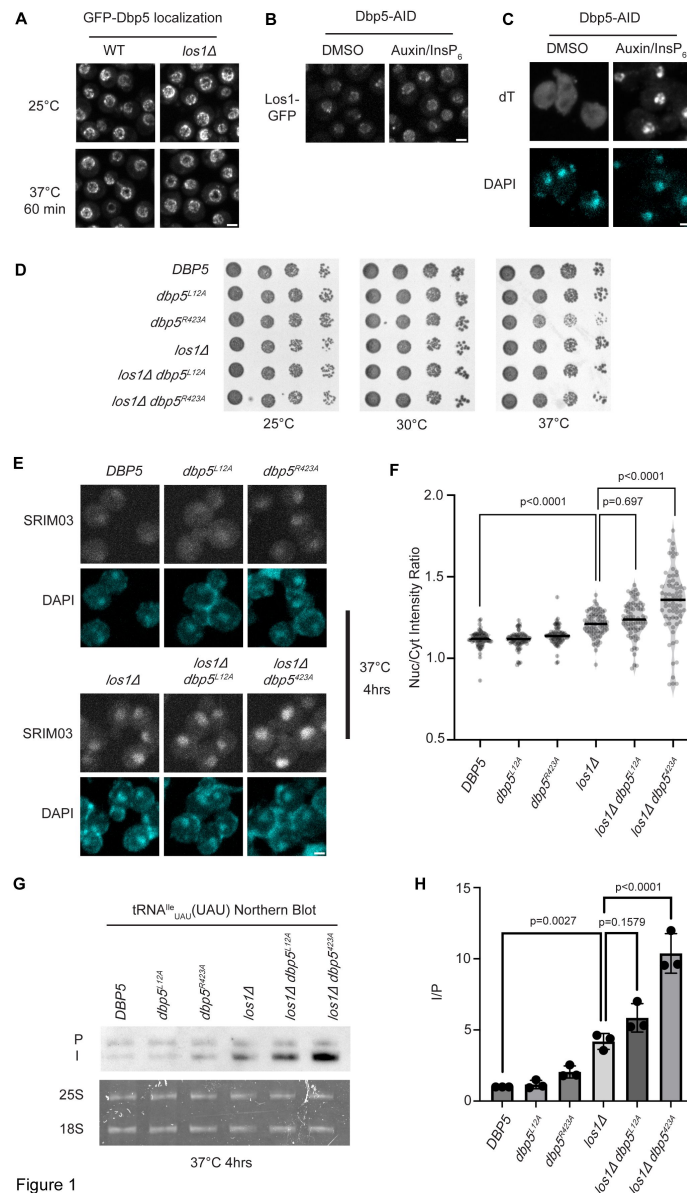


Figure 1

Figure 1

Dbp5 functions parallel to Los1 in pre-tRNA^{Ile}_{UAU} export

(A) Fluorescent images show GFP-Dbp5 remains enriched at the nuclear periphery in wild-type and *los1Δ* at 25°C and 37°C. Scale bar represents 2 μm.

(B) Los1-GFP remains nucleoplasmic and associated with nuclear periphery in Dbp5-AID after treatment with DMSO or 500 μM Auxin and 10 μM InsP₆ for 90 minutes. Scale bar represents 2 μm.

(C) dT FISH confirms induction of mRNA export defect and Dbp5 loss of function after addition of 500 μM Auxin and 10 μM InsP₆ in Dbp5-AID strain for 90 minutes. Scale bar represents 2 μm.

(D) Spot assay for growth of strains containing untagged *dbp5^{L12A}*, *dbp5^{R423A}*, or *los1Δ* integrated at the endogenous gene locus after two days at 25, 30, and 37°C on YPD.

(E) tRNA FISH targeting intron of tRNA^{Ile}_{UAU} in indicated strains after pre-culture to early log phase at 25°C and shift to 37°C for 4 hours. Scale bar represents 2 μm.

(F) Quantification of tRNA FISH from (E). Ratio of average nuclear to cytoplasmic pixel intensities were calculated across 3 independent replicate experiments and pooled for plotting. P-values were calculated using one-way ANOVA.

(G) Northern Blot analysis targeting precursor and mature isoforms of tRNA^{Ile}_{UAU}. Small RNAs were isolated from strains at mid log phase growth after pre-culture at 25°C and shift to 37°C for 4 hours.

(H) Quantification of Northern blot from (G). Ratio of signal from intron-containing end processed intermediates (I) vs 5' leader/3' trailer containing precursor (P) was calculated and presented relative to I/P ratio observed for WT. Error bars represent standard deviation and p-values calculated using one-way ANOVA.

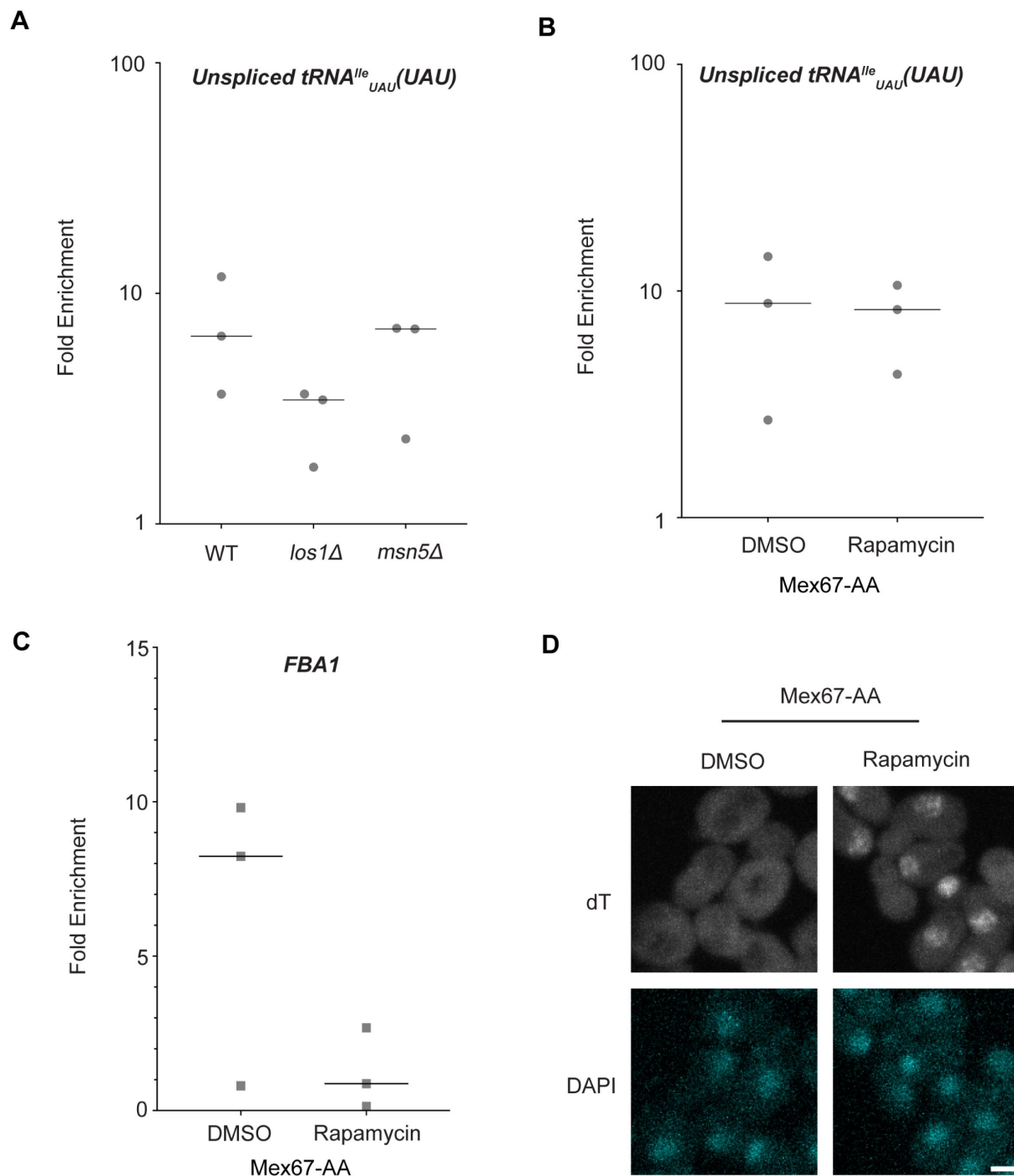


Figure 2

Figure 2

Los1 and Mex67 are not required for Dbp5 recruitment to pre-tRNA^{Ile}_{UAU}

(A) Plots show relative fold enrichment of tRNA^{Ile}_{UAU} following prA-Dbp5 RNA IP in a wild-type (WT), *los1Δ*, and *msn5Δ* strain background. Abundance of target gene is normalized to abundance of the target transcript in input samples and represented as fold enrichment relative to RNA IP from an untagged control.

(B) prA-Dbp5 RNA IP targeting tRNA^{Ile}_{UAU} in Mex67-AA after either treatment with DMSO or 1 μg/ml rapamycin for 15 minutes.

(C) prA-Dbp5 RNA IP targeting *FBA1* mRNA in Mex67-AA after either treatment with DMSO or 1 μg/ml rapamycin for 15 minutes.

(D) dT FISH confirming a mRNA export defect caused by loss of function of Mex67 following 15min incubation with 1 μg/ml rapamycin compared to a DMSO control. Scale bar represents 2 μm.

In mRNA export, Mex67 is a proposed target of Dbp5 activity to promote directional nuclear export (Adams and Wente 2020 [DOI](#)). Since neither of the non-essential tRNA export factors abolished Dbp5-tRNA interaction when deleted, it was tested if Mex67 has a role in recruiting Dbp5 to tRNA. An Anchors Away approach was used to rapidly re-localize Mex67 to the cytoplasmic peroxisomal anchor (Pex25-FKBP12) in an inducible manner dependent on the addition of rapamycin (Haruki, Nishikawa et al. 2008). The resulting strains were sensitive to rapamycin, causing nuclear mRNA accumulation by dT FISH within 15 minutes of addition (**Figure 2D** [DOI](#)). This timing is consistent with a previous report employing Mex67 Anchors Away (Mex67-AA) (Tudek, Schmid et al. 2018). PrA-Dbp5 was integrated at its endogenous locus in these strains and RNA-IP experiments were performed before and after rapamycin addition as described above. Under these conditions of Mex67-AA re-localization, no significant change in the Dbp5-tRNA^{Ile_{UAU}} interaction was observed (median of ~8.8-fold enrichment before addition of rapamycin and ~8.2-fold after, **Figure 2B** [DOI](#)). Importantly, consistent with the function of Mex67 mRNA export (Snay-Hodge, Colot et al. 1998, Tseng, Weaver et al. 1998, Schmitt, von Kobbe et al. 1999, Strahm, Fahrenkrog et al. 1999, Weirich, Erzberger et al. 2004, Lund and Guthrie 2005 [DOI](#), Tran, Zhou et al. 2007, Dossani, Weirich et al. 2009, von Moeller, Basquin et al. 2009, Adams and Wente 2020 [DOI](#)), the nuclear retention of mRNAs via Mex67-AA resulted in strongly reduced binding of Dbp5 to the *FBA1* mRNA (from a median of ~8 fold enrichment above background before addition of rapamycin to no enrichment after, **Figure 2C** [DOI](#)). Together, these RNA IP experiments suggest that Dbp5 binds to tRNAs independent of known tRNA export proteins and does so in a manner that is distinct from mRNA.

The Dbp5 ATPase cycle supports tRNA export in vivo

In mRNA export, RNA binding, Gle1 activation, and ATP hydrolysis by Dbp5 are all critical for directional transport (Hodge, Tran et al. 2011, Noble, Tran et al. 2011). The *dbp5*^{R426Q} and *dbp5*^{R369G} mutants have previously been shown to exhibit deficient RNA binding (<5% of WT), ATPase activity (~10% and 60% of WT, respectively) and have a dominant-negative effects on mRNA export status when overexpressed (Hodge, Tran et al. 2011, Noble, Tran et al. 2011). Additionally, it has been shown that *dbp5*^{R369G} has elevated Gle1 affinity and effectively competes with WT Dbp5 for Gle1 binding (Hodge, Tran et al. 2011). In contrast, *dbp5*^{E240Q} is an ATP hydrolysis mutant that is competent to bind RNA, as it is biased to the ATP bound state, but is recessive lethal showing no mRNA export defect when overexpressed in the context of endogenously expressed WT Dbp5 (Hodge, Tran et al. 2011). Unlike the impact of these mutants on mRNA export, it has been reported that the ability of Dbp5 to complete ATP hydrolysis (but not RNA binding) is dispensable for pre-ribosomal subunit export, as is Gle1 (Neumann, Wu et al. 2016).

Based on the observation that Gle1 functions to support tRNA export (Lari, Arul Nambi Rajan et al. 2019), these Dbp5 mutants deficient in ATPase activity or altered Gle1 binding were tested for their influence on tRNA export to infer if tRNA export is more similar to mRNA or pre-ribosomal subunit export. To individually express these lethal mutants and the wildtype control, *DBP5*, *dbp5*^{R426Q}, *dbp5*^{R369G}, or *dbp5*^{E240Q} were integrated at the *URA3* locus under regulation of the inducible pGAL promoter. Protein expression was induced for 6 hours by shifting cells from raffinose to galactose containing media. Importantly, induction was followed by 1 hour of growth in glucose to halt expression and relieve potential changes in tRNA export induced by a change in the primary carbon source. Using this approach, all proteins were expressed as indicated by western blotting (**Figure 3A** [DOI](#)) and showed the previously reported impact on mRNA export (**Figure 3B** [DOI](#)). Northern blotting targeting tRNA^{Ile_{UAU}} revealed both *dbp5*^{R426Q} and *dbp5*^{R369G}, but not *dbp5*^{E240Q}, showed an accumulation of intron-containing precursor tRNA^{Ile_{UAU}} as compared to over expression of wildtype Dbp5 (I/P ratio of 1.97 +/-0.06 vs 1.93 +/-0.31 vs 1.32 +/-0.41, respectively) (**Figure 3C** [DOI](#)/D). These phenotypes differ from the reported impacts on pre-ribosomal subunit transport, indicating Dbp5 functions differently in these non-coding RNA

export pathways. Furthermore, these data indicate that both the ATPase cycle and regulation of Dbp5 by Gle1 are central to the function of Dbp5 in tRNA export, as each activity is in mRNA export.

Dbp5 ATPase activity in the presence of tRNA is Gle1-dependent

Given the impact of *dbp5*^{R426Q} and *dbp5*^{R369G} overexpression on tRNA export *in vivo*, the Dbp5 ATPase cycle and role of Gle1 in the context of tRNA was further investigated *in vitro*. Previous studies have characterized Dbp5 binding to single stranded RNA (ssRNA) substrates and demonstrated Dbp5 binding to ssRNA is linked to the nucleotide state of the enzyme, with the highest affinity for ssRNA occurring when Dbp5 is ATP bound (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011, Arul Nambi Rajan and Montpetit 2021). Following ssRNA binding, hydrolysis of ATP to ADP results in a reduction in the affinity of Dbp5 for the ssRNA substrate (Weirich, Erzberger et al. 2006). While there has been extensive characterization of the structural and biochemical details of Dbp5, ssRNA, and co-regulators (Wong, Gray et al. 2018, Mason and Wentz 2020 [DOI](#), Gray, Cao et al. 2022), no such understanding exists for Dbp5 engaging highly structured RNA substrates like tRNA. As such, it was first tested if recombinant full length Dbp5 could bind commercially available yeast mixed tRNAs or the yeast Phenylalanine tRNA (both substrates that have been used in previous biochemical and structural studies (Shi and Moore 2000 [DOI](#), Yao, Roser et al. 2007)). To test whether Dbp5 can form complexes with tRNA *in vitro*, electrophoretic mobility shift assays (EMSA) were performed in the presence of different nucleotide states and tRNA substrates. Consistent with published observations of complex formation between Dbp5 and ssRNA (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011), Dbp5 generated band shifts for both mixed tRNA substrates and Phe tRNA in the presence of the ATP state analog ADP•BeF₃ (**Figure 4A** [DOI](#)). In the presence of ATP or ADP, or with no nucleotide, Dbp5 failed to form a similar band shift, which exactly parallels reported interactions between Dbp5 and ssRNA in these nucleotide states (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011).

To further estimate an affinity for tRNA, both the Phenylalanine tRNA and yeast mixed tRNAs were used in EMSA experiments with varying concentrations of Dbp5 (and fixed concentrations of ADP•BeF₃ and tRNA). From quantification of the EMSAs, an estimated K_d of ~150nM (Phe tRNA) and ~130nM (mixed tRNA) was obtained (**Figure 4B** [DOI](#)/C). In addition to EMSAs, Dbp5 binding to tRNA was observed and tested in competition assays using fluorescence polarization measurements. In these assays, increasing concentrations of unlabeled tRNA or a ssRNA homopolymeric polyadenylic acid (pA, routinely used for RNA activation in steady-state ATPase assays (Weirich, Erzberger et al. 2006)), were found to compete with a fluorescently labeled ssRNA for Dbp5 in the presence of ADP•BeF₃ (**Figure 4D** [DOI](#)). These data, from orthogonal approaches, indicate that Dbp5 interacts directly with tRNA *in vitro*, and does so in a manner governed by principles that resemble Dbp5-ssRNA binding.

Interaction with ssRNA substrates has been well documented to stimulate the Dbp5 ATPase cycle (Weirich, Erzberger et al. 2004, Lund and Guthrie 2005 [DOI](#), Alcazar-Roman, Tran et al. 2006 [DOI](#), Weirich, Erzberger et al. 2006, Tran, Zhou et al. 2007, Dossani, Weirich et al. 2009, von Moeller, Basquin et al. 2009, Hodge, Tran et al. 2011, Montpetit, Thomsen et al. 2011, Noble, Tran et al. 2011). As such, it was next determined if binding of tRNA to Dbp5 stimulates ATPase activity using a spectrophotometric ATPase assay (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011). Unlike a ssRNA substrate (i.e., pA) that can maximally stimulate ATP turnover to ~0.5 ATP/sec, neither tRNA nor poly(I:C) (used as an orthogonal dsRNA substrate) stimulated Dbp5 ATPase activity over a range of RNA concentrations (**Figure 5A** [DOI](#)). These results with the above binding data suggest Dbp5 can engage structured and dsRNA substrates, but this does not lead to productive ATP hydrolysis.

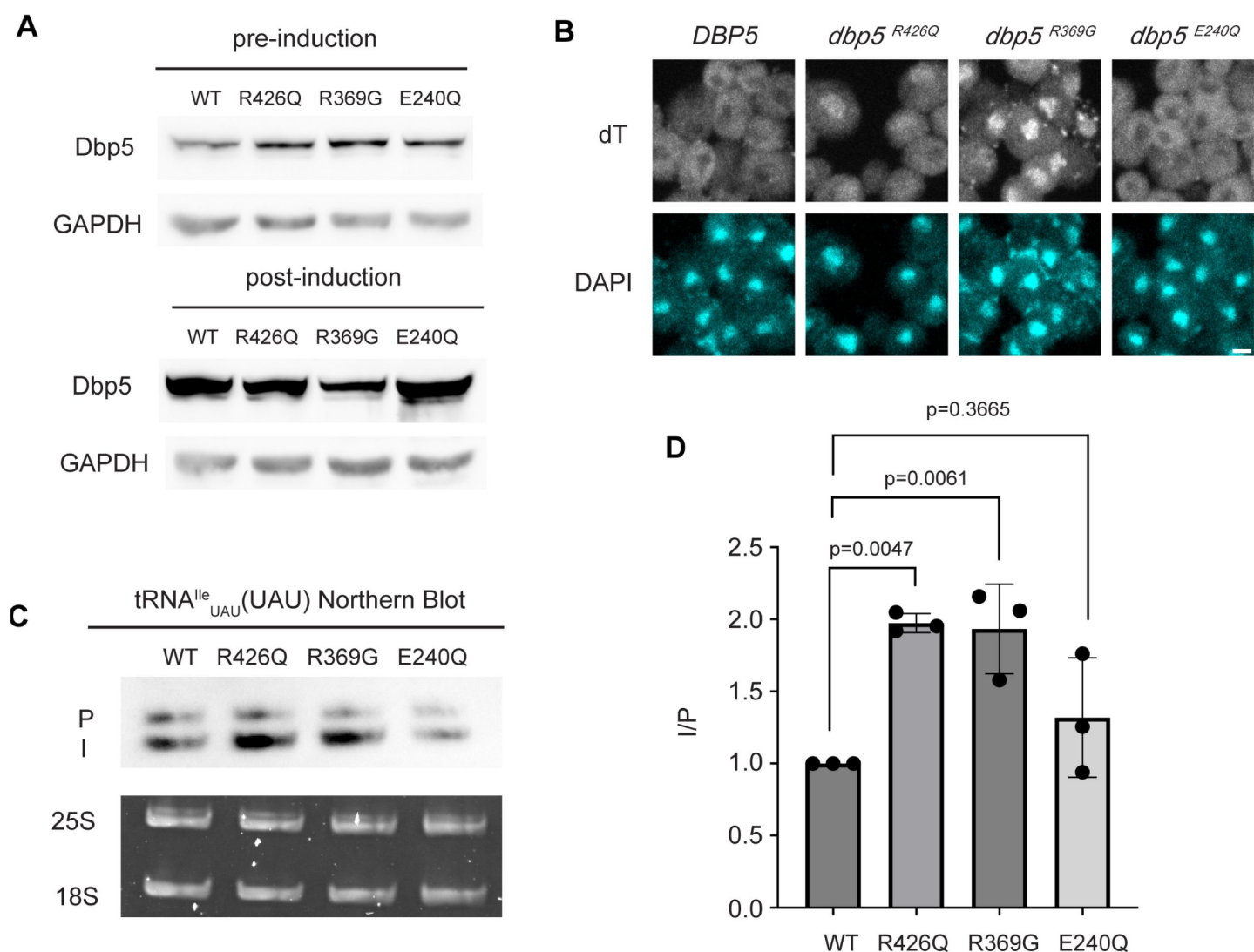


Figure 3

Figure 3

ATPase activity and Gle1 is required for Dbp5-mediated tRNA export in vivo

(A) Western blot with either mouse monoclonal anti-DBP5 or anti-GAPDH (loading control) antibody to detect overexpression of untagged Dbp5 and Dbp5 ATPase mutants. Strains were cultured in raffinose at 25°C to derepress pGAL promoter and expression was induced by shifting cultures into 2% galactose containing media for 6 hours. Expression was stopped by addition of glucose for 1 hour. Overexpression was observed over the level of wildtype Dbp5 that is expressed from the endogenous locus and is present in the pre-induction samples. Top and bottom panels are from the same blot and processed equally.

(B) dT FISH confirms previously reported mRNA export status phenotypes for Dbp5 ATPase mutants with *dbp5*^{R426Q} and *dbp5*^{R369G} showing a nuclear accumulation of poly(A)-RNA. Scale bar represents 2 μm.

(C) Northern blot analysis targeting precursor and mature isoforms of tRNA^{Ile}_{UAU} from yeast strains overexpressing DBP5, *dbp5*^{R426Q}, *dbp5*^{R369G}, or *dbp5*^{E240Q}.

(D) Quantification of Northern blot from (C). Ratio of signal from intron-containing end processed intermediates (I) vs 5' leader/3' trailer containing precursor (P) was calculated and presented relative to I/P ratio observed for WT. Error bars represent standard deviation and p-values calculated using one-way ANOVA.

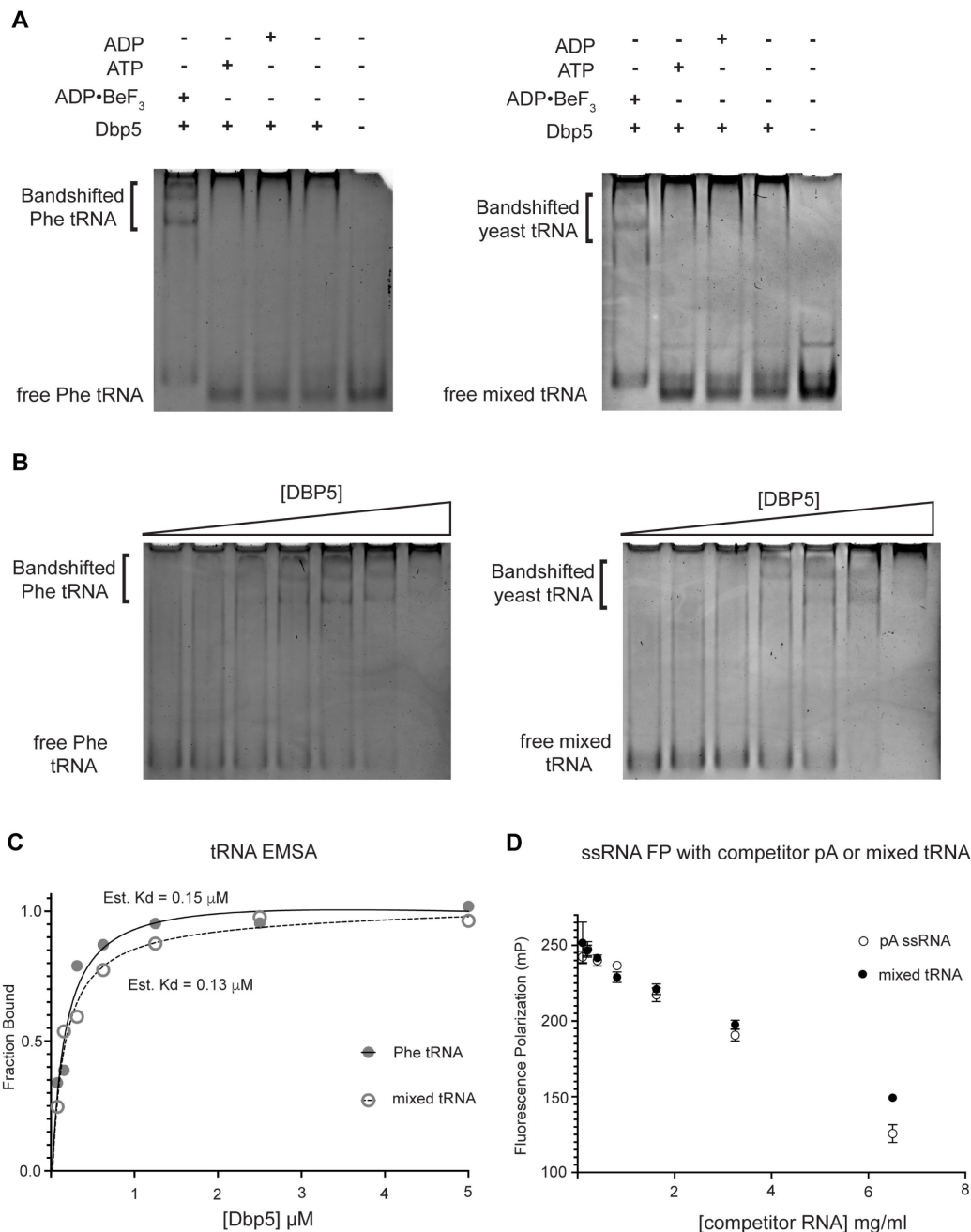


Figure 4

Figure 4

Dbp5 binds yeast tRNA in vitro

(A) Full length recombinant Dbp5 (fl-Dbp5) binds both mixed yeast tRNA and Phenylalanine (Phe) tRNA in the presence of the ATP mimetic ADP•BeF₃ by electrophoresis mobility shift assay (EMSA). RNA binding reaction was conducted in the presence of ADP•BeF₃, ATP, ADP, or no nucleotide and resolved on 6% native polyacrylamide gel at 70V. Reactions contained 2 μM fl-Dbp5, 1 mM nucleotide when present, 250ng tRNA and binding buffer.

(B) EMSA experiments in which increasing concentrations of fl-Dbp5 was titrated into RNA binding reactions in the presence of 1 mM ADP•BeF₃, 250ng mixed yeast tRNA or Phe tRNA, and binding buffer.

(C) Band intensities of free probe and band shifts were quantified, and bound fraction was calculated for each well of EMSAs in (B). One site binding model was fit to the data using GraphPad Prism to estimate K_d for both mixed yeast tRNA and Phe tRNA.

(D) Unlabeled pA ssRNA and mixed yeast tRNA compete with a 16nt fluorescein labelled ssRNA for fl-Dbp5 binding. Fluorescence polarization competition assays were performed by titrating increasing concentration of an unlabeled competitor, pA (open square) or mixed yeast tRNA (closed circle), in reactions containing 50 nM fluorescein labelled ssRNA, 2.5 mM ADP•BeF₃, 1 μM Dbp5 and buffer. Error bars represent standard deviation of three independent experiments.

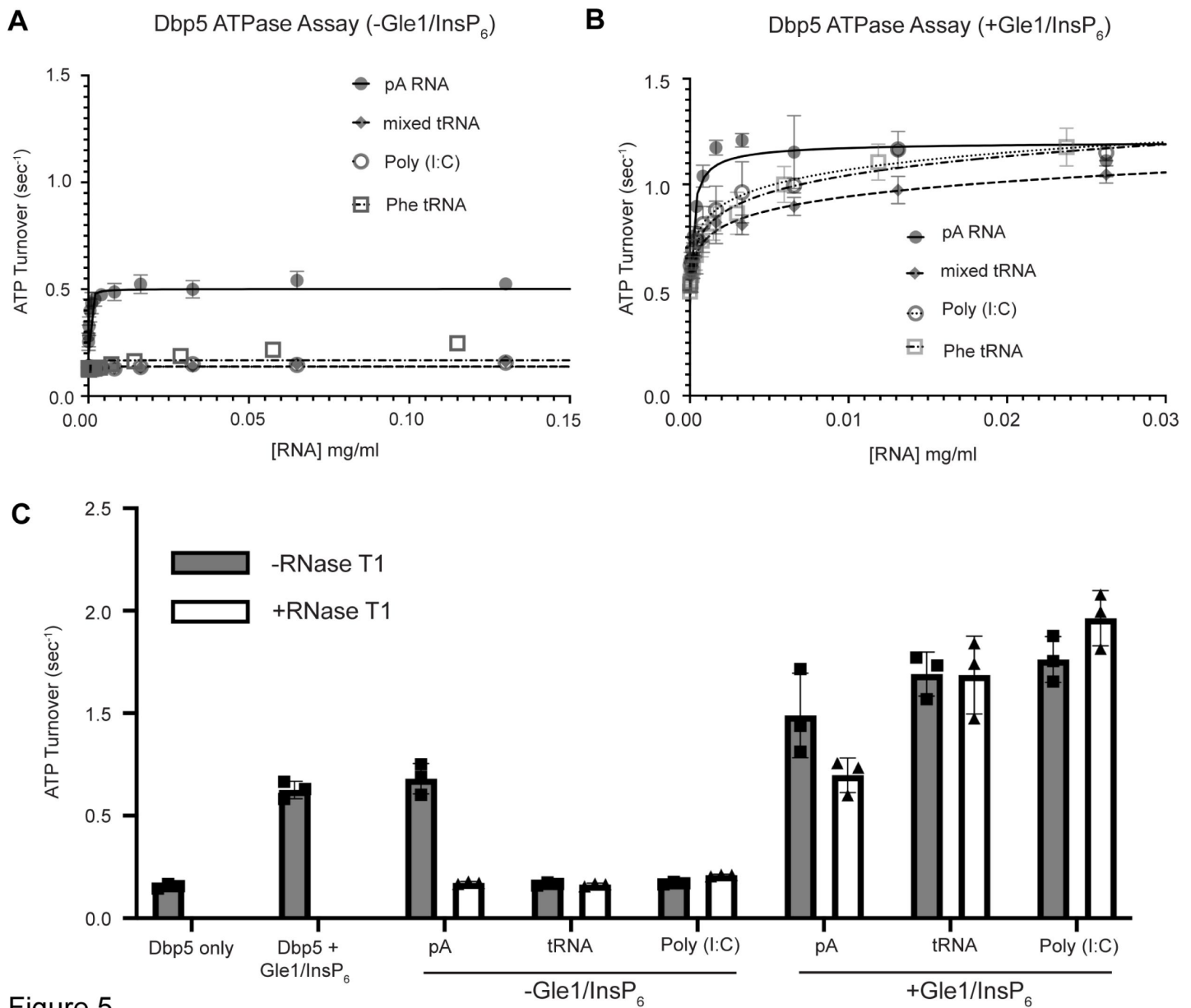


Figure 5

Figure 5

tRNA alone does not stimulate the Dbp5 ATPase cycle but can act synergistically with Gle1/InsP₆ to fully activate Dbp5

(A) ATPase activity of Dbp5 is stimulated by ssRNA (pA, closed square) but not mixed yeast tRNA (closed diamond), Phe tRNA (open square), or poly (I:C) dsRNA (open circle). Steady-state ATPase assays were conducted in the presence of 1 μ M Dbp5, 2.5 mM ATP, and varying concentrations of RNA. Data was fit to Michaelis-Menten equation using GraphPad Prism. Error bars represent standard deviation of three independent experiments.

(B) Gle1/InsP₆ synergistically stimulates Dbp5 ATPase activity with mixed yeast tRNA (closed diamond), Phe tRNA (open square) and poly (I:C) dsRNA (open circle) substrates like ssRNA (pA). Steady-state ATPase assays were conducted in presence of 1 μ M Dbp5, 2.5 mM ATP, 2 μ M Gle1, 2 μ M InsP₆, and varying concentration of RNA. Data was fit to an allosteric sigmoidal model in GraphPad Prism. Error bars represent standard deviation of three independent experiments.

(C) RNase T1 treatment of RNA for 2 hours at 37°C prior to ATPase assays confirms that the observed synergistic activation of Dbp5 ATPase activity by Gle1/InsP₆ and tRNA or dsRNA is not caused by low levels of contaminating ssRNA. Steady-state ATPase assays were conducted in presence of 1 μ M Dbp5, 2.5 mM ATP, 0.2 mg/ml RNA and 2 μ M Gle1/InsP₆ when indicated.

The nucleoporin Gle1 with the small molecule inositol hexakisphosphate (InsP₆) has been shown to synergistically activate Dbp5 ATPase activity at low RNA concentrations (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011). These findings have led to a model of spatially regulated Dbp5 activity to promote mRNA-protein complex remodeling at the cytoplasmic face of an NPC where Gle1 is localized, resulting in directional mRNA export (Hodge, Colot et al. 1999, Strahm, Fahrenkrog et al. 1999, Alcazar-Roman, Tran et al. 2006 [↗](#), Weirich, Erzberger et al. 2006, Dossani, Weirich et al. 2009, Hodge, Tran et al. 2011, Montpetit, Thomsen et al. 2011, Noble, Tran et al. 2011, Adams, Mason et al. 2017 [↗](#), Wong, Gray et al. 2018, Arul Nambi Rajan and Montpetit 2021). In the context of tRNA, addition of Gle1/InsP₆ maximally stimulated Dbp5 (1.03 +/-0.04 ATP/s) to a level like that of ssRNA (1.11 +/-0.07) (**Figure 5B** [↗](#)). The synergistic activation of Dbp5 ATPase activity by Gle1/InsP₆ and tRNA is mirrored by poly(I:C) and observed with both mixed yeast tRNAs as well as yeast Phenylalanine tRNA (**Figure 5B** [↗](#)). To confirm that this enhanced RNA activation by Gle1/InsP₆ was not the result of contaminating ssRNA, ATPase assays were performed after treatment of tRNA or poly(I:C) with RNase T1 for 2 hours at 37°C. RNase T1 degrades ssRNA, not dsRNA, allowing determination of whether RNA activation observed in the presence of Gle1/InsP₆ is specific to properly folded tRNA and poly(I:C). As expected, treatment of the ssRNA (pA) with RNase T1 resulted in the loss of RNA-stimulated Dbp5 ATPase activity (0.68 +/-0.06 before RNase T1 treatment to 0.17 +/-0.01 after). Furthermore, RNase T1 treatment reduced pA/Gle1/InsP₆ activation (0.98 +/-0.17 before treatment to 0.69 +/-0.07 after) to that of Dbp5 and Gle1/InsP₆ alone (0.62 +/-0.03) (**Figure 5C** [↗](#)). However, RNase T1 treated tRNA remained unchanged in the ability to activate Dbp5 in the presence of Gle1/InsP₆ (1.19+0.11 without vs 1.18+/-0.19 ATP/s with RNase T1 treatment). This was recapitulated with poly(I:C) (1.26 +/-0.11 ATP/s without and 1.46 +/-0.135 ATP/s with RNase T1), confirming that the synergistic activation of Dbp5 observed in the presence of Gle1/InsP₆ is dependent on the structured tRNA or dsRNA. These findings provide for the possibility that Dbp5 engages tRNA in the cell without ATPase activation and could remain bound until the Dbp5-tRNA complex reaches Gle1 at the cytoplasmic side of a nuclear pore complex.

Discussion

The best characterized tRNA export factors in *S. cerevisiae*, Los1 and Msn5, are non-essential and the *los1Δ/msn5Δ* double mutant is viable despite the essential role of tRNA export (Takano, Endo et al. 2005, Murthi, Shaheen et al. 2010). These data suggest additional mechanisms for regulated tRNA export, which is supported by recent publications that have implicated mRNA and rRNA export factors such as Mex67, Nup159, Gle1, Dbp5, and Crm1 in tRNA export (Wu, Bao et al. 2015, Chatterjee, Majumder et al. 2017, Nostramo and Hopper 2020 [↗](#), Chatterjee, Marshall et al. 2022). However, a mechanistic understanding of how these factors function in tRNA export and how such roles differ or relate to previously characterized roles in RNA export is still unresolved. Here, a combination of *in vivo* and *in vitro* data provide evidence that (1) Dbp5 functions parallel to Los1 in tRNA export, similar to the mRNA export factor Mex67 and Crm1; (2) Dbp5 can bind tRNA directly *in vitro* and does so independently of Los1, Msn5, and Mex67 *in vivo*; and (3) the Dbp5 ATPase cycle is uniquely modulated by Gle1 in the presence of structured RNA (e.g. tRNA or dsRNA) *in vitro* and the ATPase cycle supports tRNA export *in vivo*.

Collectively, the *in vitro* characterization of Dbp5 and tRNA described here suggest two non-mutually exclusive possibilities for why Dbp5 is only activated by tRNA/dsRNA in the presence of Gle1/InsP₆. First, it may be that Dbp5 binds tRNA to form an inhibited intermediate that is relieved by Gle1/InsP₆. This is an attractive hypothesis that is unique from how Dbp5 engages ssRNA (e.g., mRNA) and potentially supports previously proposed export models that suggest Dbp5 may bind and travel with a tRNA from nucleus to cytoplasm (Lari, Arul Nambi Rajan et al. 2019). A second possibility is that Gle1 enhances the affinity of Dbp5 for tRNA and “non-mRNA” substrates, which is supported by biochemical characterizations that show Gle1 promotes a shift in the steady-state distribution of populated Dbp5 intermediates from weak to strong RNA binding states (Hodge, Tran et al. 2011, Montpetit, Thomsen et al. 2011, Noble, Tran et al. 2011, Wong, Cao et al. 2016,

Wong, Gray et al. 2018, Gray, Cao et al. 2022). Future structural and biochemical studies are expected to distinguish among these possibilities. Moreover, Dbp5 was previously reported to not bind dsRNA substrates (Weirich, Erzberger et al. 2006), as such the findings that Dbp5 can bind, and in the presence of Gle1, be activated by tRNA should motivate reevaluation of possible functions of Dbp5-Gle1/InsP₆ in regulating other highly structured RNA or “non-canonical” substrates *in vivo*.

While it has been reported previously that tRNA fails to stimulate Dbp5 ATPase cycle (Tseng, Weaver et al. 1998), here it is shown that despite a lack of stimulation when Gle1/InsP₆ is absent, Dbp5 is able to bind tRNA *in vitro* and *in vivo*. The ability of a tRNA to bind but not stimulate the ATPase cycle of Dbp5 has important implications for mechanisms by which Dbp5 may function in tRNA export. For example, in mRNA export it is thought that Dbp5 functions are localized at NPCs (Derrer, Mancini et al. 2019, Lari, Arul Nambi Rajan et al. 2019, Adams and Wentz 2020 [DOI](#)), where the Dbp5 ATPase cycle is tightly regulated by co-factors Nup159 and Gle1/InsP₆ (Hodge, Colot et al. 1999, Rollenhagen, Hodge et al. 2004, Hodge, Tran et al. 2011, Montpetit, Thomsen et al. 2011, Noble, Tran et al. 2011, Wong, Gray et al. 2018). In fact, recent studies have shown the essential function of both Dbp5 and Mex67 in mRNA export can be accomplished when these proteins are fused to nuclear pore components, in addition Dbp5 does not form a complex with mRNAs in the nucleus (Derrer, Mancini et al. 2019, Adams and Wentz 2020 [DOI](#)). These reports suggest that Dbp5 does not travel with an mRNA from the nucleoplasm to the cytoplasmic face of an NPC as part of an mRNA-protein complex, rather Dbp5 likely engages mRNAs after they exit the NPC transport channel to direct the final step(s) of mRNA export (**Figure 6** [DOI](#)). However, in contrast to this mRNA export model, the unique ability of Dbp5 to bind tRNA without ATP stimulation, and then be fully activated by Gle1/InsP₆, presents the potential for a novel mechanism of function in regulating tRNA export. Namely, that Dbp5 may enter the nucleus to stably engage tRNA and direct events leading to export, following which Dbp5 is removed from the tRNA by Gle1/InsP₆ upon nuclear exit (**Figure 6** [DOI](#)). Importantly, in both mRNA and tRNA export, Gle1/InsP₆ activation is required to stimulate ATPase activity and recycle the enzyme. In contrast, for pre-ribosomal subunit export Dbp5 ATPase activity and Gle1 have been shown to be dispensable for transport (Neumann, Wu et al. 2016). This raises interesting questions of whether RNA binding and nucleocytoplasmic shuttling of Dbp5 alone can achieve this function, where in the cell these interactions occur, and how rRNA substrates modulate Dbp5 activity.

In mRNA export, it has been proposed that Dbp5 displaces mRNA export factors such as Mex67 and Nab2 to promote directionality of mRNA transit (Lund and Guthrie 2005 [DOI](#), Tran, Zhou et al. 2007). Observations that Mex67 and Dbp5 appear to both function parallel to Los1 to promote tRNA export may point to an overlapping pathway that shares both mRNA export factors. Dbp5 and Mex67 have been proposed to form physical interactions, with Mex67 serving as an adaptor to direct Dbp5 to appropriate mRNA sites for remodeling (Lund and Guthrie 2005 [DOI](#)). However, here it is demonstrated that Mex67 is not required for Dbp5-tRNA interaction based on RNA IP assays. Additionally, previously published data suggests Mex67 likely requires an unknown adaptor to form its reported interactions with its tRNA substrates (Yao, Roser et al. 2007). Given the *in vitro* properties of Dbp5 binding to tRNA, it is tempting to speculate that nuclear Dbp5 promotes recruitment of Mex67 or Crm1 to tRNA-protein complexes. Indeed, a similar phenomenon has been described for the exon junction complex (EJC) and the DEAD-box protein at its core, eIF4AIII, that acts as a platform for the binding of other proteins to an mRNA (Ballut, Marchadier et al. 2005, Andersen, Ballut et al. 2006, Bono, Ebert et al. 2006, Pacheco-Fiallos, Vorlander et al. 2023). Stable binding of the EJC is governed by trans acting proteins that lock the complex on the mRNA by stabilizing an ADP-P_i hydrolysis intermediate (Ballut, Marchadier et al. 2005, Bono, Ebert et al. 2006, Nielsen, Chamieh et al. 2009, Linder and Fuller-Pace 2013 [DOI](#)). In the case of Dbp5, the *in vitro* data suggests an inhibited state may be achieved in the context of tRNA alone, which could be further stabilized *in vivo* by yet to be discovered protein factors. The ability of Gle1/InsP₆ to induce maximal activity of Dbp5 in the presence of tRNA would then allow for resolution of stable Dbp5-tRNA complexes in a spatially regulated manner to terminate export and recycle bound export

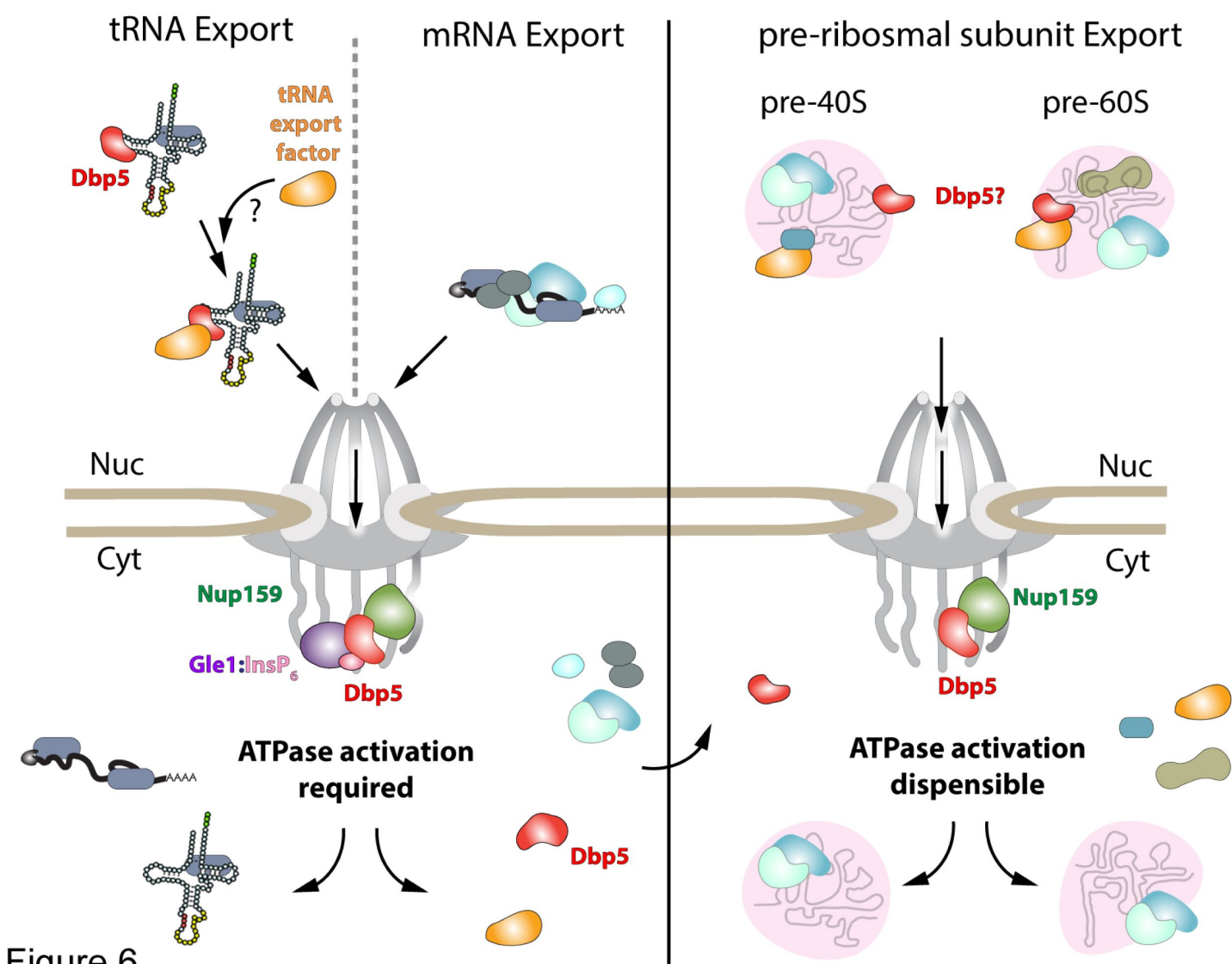


Figure 6

Figure 6

Model of Dbp5 function in mRNA, tRNA and pre-ribosomal subunit export

Data from this study support a model in which Dbp5-mediated export of both mRNA and tRNA require Gle1/InsP₆ activation to remodel RNPs at the cytoplasmic face of the Nuclear Pore Complex. For tRNA export, lack of RNA-mediated ATPase stimulation following RNA binding may lead to formation of tRNA bound intermediates that act as adapters for recruitment of yet to be identified transport factors. For mRNA export, it has been shown that Dbp5 function is limited to the nuclear periphery and Dbp5 does not form stable nuclear complexes with mRNA. Gle1/InsP₆ mediated activation of Dbp5 catalytic cycle and RNA release likely promotes recycling of export factors and Dbp5 to function in further rounds of export or other functions. In contrast, for Dbp5 mediated pre-ribosomal subunit export Dbp5 ATPase cycle and Gle1/InsP₆ stimulation are dispensable for transport. As such, RNA binding may promote export through an unknown mechanism and that Dbp5-mediated remodeling does not occur at NPCs, with factors such as Mex67 persisting on pre-ribosomal subunits following export to the cytoplasm.

factors (**Figure 6** [↗](#)). Alternatively, given that it has been reported that Mex67 can bind 5' extended tRNAs and that these transcripts have been previously reported to contain 5' methyl-guanosine cap structures (Ohira and Suzuki 2016 [↗](#), Chatterjee, Marshall et al. 2022), perhaps an “mRNA export-like” pathway that involves Cap Binding Complex (CBC) can act to recruit Mex67 to a subset of tRNAs. It remains unclear if CBC interacts with such transcripts and whether other export factors like Dbp5 also support “premature” export of unprocessed tRNAs.

More broadly, like many DEAD-box proteins, Dbp5 has many reported roles in controlling gene expression (Estruch and Cole 2003 [↗](#), Rollenhagen, Hodge et al. 2004, Gross, Siepmann et al. 2007, Scarcelli, Viggiano et al. 2008, Tieg and Krebber 2013 [↗](#), Wu, Becker et al. 2014, Neumann, Wu et al. 2016, Mikhailova, Shuvalova et al. 2017, Lari, Arul Nambi Rajan et al. 2019, Beissel, Grosse et al. 2020). The nucleic acid substrate-specific biochemical properties of Dbp5 ATPase regulation defined here may have wide-ranging implications for Dbp5 functions in pre-ribosomal RNA export, translation, or R-loop metabolism. Furthermore, several DEAD-box proteins such as Dbp2 have been reported to bind diverse nucleic acid substrates (ncRNA and mRNA), including G-Quadruplexes, and have roles in R-loop biology (Ma, Paudel et al. 2016, Xing, Wang et al. 2017, Tedeschi, Cloutier et al. 2018, Gao, Byrd et al. 2019, Lai, Choudhary et al. 2019, Lai and Tran 2021 [↗](#), Yan, Obi et al. 2021, Song, Lai et al. 2022). Given the high degree of conservation in the structure, function, and regulation of Dbp5 with other DEAD-box proteins (Ozgur, Buchwald et al. 2015, Sloan and Bohnsack 2018 [↗](#)), the observations reported here raise the possibility that other DEAD-box proteins may also have substrate-specific regulation.

Overall, this study and other recent publications support a general function for both Dbp5 and Mex67 in RNA export (Yao, Roser et al. 2007, Yao, Lutzmann et al. 2008, Faza, Chang et al. 2012, Wu, Becker et al. 2014, Neumann, Wu et al. 2016, Chatterjee, Majumder et al. 2017, Becker, Hirsch et al. 2019, Lari, Arul Nambi Rajan et al. 2019, Nostramo and Hopper 2020 [↗](#), Vasianovich, Bajon et al. 2020, Chatterjee, Marshall et al. 2022). While these factors have been most well characterized in mRNA export, it is now important to reconsider Dbp5 and Mex67 as general RNA export factors, which raises questions about possible co-regulation of mRNA and non-coding RNA export pathways to fine tune gene expression. To this end, the *in vitro*, biochemical, and genetic properties of Dbp5 reported in this study will inform future structure-function and mechanistic characterization of a Dbp5 mediated, and Gle1 regulated, tRNA export pathway(s). For example, it will be important for future studies to address the composition of pathway specific export-competent tRNA-protein complexes and how Dbp5 contributes to changes in the architecture of such a complex as they move from nucleus to cytoplasm and back again.

Materials and methods

Yeast strains generation and growth conditions

A list of all yeast strains used in this study is provided in Table S1. Deletion mutants and C-terminal tagging was conducted by PCR based transformations and confirmed by colony PCR. All yeast transformations were conducted using previously published standard high-efficiency LiAc, ssDNA, PEG protocol (Gietz and Woods 2002 [↗](#)). Yeast were cultured in YPD or synthetic complete (SC) media when indicated to mid log growth. For overexpression of Dbp5 ATPase mutants, untagged Dbp5 or *dbp5*^{R426Q}, *dbp5*^{R369G}, or *dbp5*^{E240Q} were integrated at URA3 locus under control of pGAL promoter in a wild-type strain with Dbp5 expressed from its endogenous locus and promoter. pGAL expression was induced by first derepressing the promoter with growth in YP media with 2% raffinose overnight followed by a shift to 2% galactose containing media for 6 hours. Overexpression was halted by addition of 2% glucose and further incubation for 1 hour.

Spot growth assay

Yeast were pre-cultured to saturation overnight, diluted, and 1:10 serial dilutions were made with the most concentrated wells representing a concentration of 0.25 OD/ml. 3 μ L of each dilution was spotted on YPD and grown at indicated temperatures for 2 days.

Live Cell Imaging

Imaging experiments were carried out using the confocal configuration of an Andor Dragonfly microscope equipped with an EMCCD camera driven by Fusion software (Andor) using a 60x oil immersion objective (Olympus, numerical aperture (N.A.) 1.4). Images were acquired from cells grown to mid log phase in SC media at 25°C and shifted to 37°C for indicated time or treated with DMSO or 500 μ M Auxin and 10 μ M InsP6 for relevant experiments. Cells were immobilized in 384-well glass bottomed plates (VWR) that were pre-treated with Concanavalin A.

tRNA extraction and Northern Blotting

Isolation of small RNAs and Northern blot experiments were performed as previously described (Wu, Huang et al. 2013). Briefly, small RNAs were isolated from yeast cultures grown to early log phase by addition of equal volumes cold TSE buffer (0.01M Tris pH7.5, 0.01M EDTA, 0.1M sodium chloride) and TSE saturated phenol. Samples were incubated at 55°C for 20min with vortexing every 3 min then placed on ice for 10 min. Aqueous phase was extracted after centrifugation, re-extraction with phenol was performed, and RNA was precipitated overnight in ethanol at -80°C. 2.5 μ g of RNA was then separated on 10% TBE-Urea gels, gels were stained with Apex safe DNA gel stain to visualize 25S and 18S rRNA and then transferred to Hybond N⁺ membrane (Amersham). RNA was crosslinked to membranes at 2400J/m² using UV Crosslinker (VWR) and tRNA were detected using digoxigenin-labeled (DIG) probes. Mean integrated intensity values were measured for I and P bands using FIJI (Schindelin, Arganda-Carreras et al. 2012) and normalized to background signal in each lane.

Sequence of Northern blot probe targeting precursor and mature isoforms of tRNA^{Ile}_{UAU} used in **Figure 2** and **4** is as follows:

GGCACAGAACTTCGGAACCGAATGTTGCTATAAGCACGAAGCTCTAACCACTGAGCTACACGA GC.

Fluorescence In-Situ Hybridization (FISH)

FISH experiments were carried out as described in previous publications with minor modifications (Lord, Ospovat et al. 2017) using directly Cy3 labeled tRNA probes or fluorescein isothiocyanate (FITC) labeled dT probes. Sequence of Cy3 labeled tRNA^{Ile}_{UAU} intron specific probe is same as previously published SRIM03 (Chatterjee, Majumder et al. 2017) with Cy3 moiety appended at the 5' end (CGTTGCTTTTAAAGGCCTGTTGAAAGGTCTTTGGCACAGAACTTCGGAACCGAATGTTGCTAT). Briefly, yeast cultures were grown to early log phase at 25°C and treated with drug/vehicle or shifted to 37°C for 4 hours when appropriate. Samples were fixed with 0.1 volume of 37% formaldehyde for 15min. These samples were then pelleted and resuspended in 4% paraformaldehyde (PFA) solution (4% PFA, 0.1M Potassium Phosphate (pH 6.5), 0.5mM MgCl₂) and incubated with rotation at 25°C for 3 hours. Cells were then washed twice with Buffer B (1.2M Sorbitol, 0.1M Potassium Phosphate (pH 6.5), 0.5mM MgCl₂). For spheroplasting cell pellets were resuspended in a 1ml Buffer B solution containing 0.05% beta-mercaptoethanol. 15 μ l of 20mg/mL zymolyase T20 was then added to each sample and were subsequently incubated at 37°C for 45min. After washing once with Buffer B, cells were adhered to eight well slides (Fisher Scientific) pre-coated with Poly-L-Lysine. Slides were placed sequentially in 70%, 90%, and 100% ethanol for 5min and then allowed to dry. Samples were next incubated at 37°C for 2 hours in pre-hybridization buffer (4X SSC, 50% formamide, 10% dextran sulfate, 125 μ g/mL *E. coli* tRNA, 500 μ g/mL salmon sperm DNA, 1x Denhardt's solution, and 10mM vanadyl ribonucleoside complex (VRC)). Pre-hybridization solution was then aspirated and replaced with pre-warmed pre-

hybridization solution that contained either 0.25uM Cy3 tRNA probe or 0.025uM FITC dT probe and further incubated at 37°C overnight. Wells were then sequentially washed for 5min once with 2x SSC, three times with 1xSSC, and once with 0.5x SSC. Slides were dipped in 100% ethanol, air-dried, and mounting media with 4', 6-diamidino-2-phenylidole (DAPI) was applied to each well prior to sealing of coverslip onto slide. Imaging was performed using Andor Dragonfly equipped with Andor iXon Ultra 888 EMCCD camera driven by Fusion software (Andor) using a 60x 1.4 N.A. oil objective. Images were processed in FIJI (e.g., cropping, brightness/contrast adjustments and maximum z-projections) (Schindelin, Arganda-Carreras et al. 2012). Quantification of tRNA FISH was performed by acquiring average nuclear and cytoplasmic pixel intensities respectively. Maximum projection images were generated from z stack images for DAPI and tRNA FISH channels. Nuclear and whole cell masks were then respectively generated using Cellpose software (Stringer, Wang et al. 2021, Pachitariu and Stringer 2022 [\[1\]](#)). Nuclear mask regions were removed from whole cell masks to generate cytoplasmic mask and average pixel intensities were calculated. The ratio of average nuclear and cytoplasmic mask pixel intensity per cell was calculated.

RNA Immunoprecipitation (RIP)

RIPs were conducted as previously described with minor modifications (Lari, Arul Nambi Rajan et al. 2019). Pull downs were performed targeting protein-A (prA) tagged Dbp5 in parallel with an untagged control to assess background non-specific binding. Crosslinking was conducted on yeast cultures in mid-log growth by addition of formaldehyde to final concentration of 0.3% and incubation for 30 min. Crosslinking was quenched by addition of glycine to a final concentration of 60mM for 10 min. Cells were harvested and pellets frozen in liquid nitrogen. Lysis was performed using ice cold TN150 (50mM Tris-HCl pH7.8, 150mM NaCl, 0.1% IGEPAL, 5mM beta-mercaptoethanol) supplemented with 10mM EDTA, 1ng Luciferase Spike-In RNA (Promega), and 1X protease inhibitor cocktail. 1ml zirconia beads (0.5mm) were used for disrupting cells by vortexing 5 times for 30 seconds followed by 1min on ice between each pulse. Lysate was pre-cleared by centrifugation at 4000xg 5min followed by 20min at 20000xg at 4°C. 1% pre-IP lysate was preserved as input RNA sample and additional pre-IP sample was reserved for western blotting. Remaining lysates were then diluted to 10ml with TN150 and incubated with IgG-conjugated magnetic dynabeads at 4°C for 30min with constant rotation. Immunoprecipitate was washed once with 1ml TN150, once with 1ml TN1000 (50mM Tris-HCl pH7.8, 1M NaCl 0.1% IGEPAL, 5mM beta-mercaptoethanol), and again once with 1ml TN150 each for 5min at 4°C. Beads from RIPs were then resuspended in 500ul proteinase K elution mix (50mM Tris-HCl pH 7.8, 50mM NaCl, 1mM EDTA, 0.5% SDS, 100ug proteinase K) in parallel with input samples and incubated at 50°C for 2 hours followed by 65°C for 1 hour to allow crosslink reversal. RNA was isolated by extraction with acidic Phenol:Chloroform:Isoamylalcohol (pH 4.3-4.7) and ethanol precipitation. Half volume of RNA was reverse transcribed using Superscript III using random priming according to manufacturer instructions and other half of RNA was retained for minus RT control. RNA was analyzed by qPCR using Power SYBR (Applied Biosystems) on an Applied Biosystems instrument. Target RNA abundance in RIP was normalized to abundance of target in input sample and relative fold enrichment was calculated by comparing signal in specific prA-Dbp5 RIPs to signal from untagged control. Standard curves were generated to test PCR efficiencies of each primer set used in this study. Primer sequences used are as follows:

tRNA^{Ile}_{UAU} Unspliced Forward: GCTCGTGTAGCTCAGTGGTTAG

tRNA^{Ile}_{UAU} Unspliced Reverse: CTTTAAAGGCCTGTTTGAAAG

FBA1 Forward: CGAAAACGCTGACAAGGAAG

FBA1 Reverse: TCTCAAAGCGATGTCACCAG

Western Blotting

~5 OD pellets were lysed in laemmli buffer and loaded onto 10% SDS PAGE. Protein transfer was performed onto nitrocellulose membrane using cold wet tank transfer protocol. Western blotting was performed with 1:5000 dilution of mouse monoclonal anti-DBP5 and 1:10000 dilution of mouse monoclonal anti-GAPDH (Thermo Fisher) primary antibody overnight at 4°C and 1:10000 anti-mouse DyeLight 650 secondary to detect protein.

Protein Purification, ATPase assays and Fluorescence Polarization

Protein purification, *in vitro* ATPase assays and fluorescence polarization using full-length Dbp5 performed as described previously (Weirich, Erzberger et al. 2006, Montpetit, Thomsen et al. 2011, Montpetit, Seeliger et al. 2012). Commercially available Poly (I:C), mixed yeast tRNA and yeast Phenylalanine tRNA were obtained from Sigma for use in indicated assays. For RNase T1 ATPase assays, indicated RNA samples were treated with 33 units of RNase T1 for 2 hours at 37°C prior to experiment. Fluorescence polarization experiments were performed using a fluorescein (fl) labelled 16nt ssRNA (5'-fl-GGGUAAAAAAAAAAAAA-3') in the presence of ADP•BeF₃ with increasing concentrations of unlabeled polyadenylic acid or yeast mixed tRNA titrated into the reactions. Fluorescence polarization and ATPase assays were assembled in the same ATPase buffer (30mM HEPES [pH 7.5], 100mM NaCl, and 2mM MgCl₂). All ATPase assays with RNA titrations were fit to Michaelis Menten or allosteric sigmoidal models (when Gle1/InsP₆ was added) in GraphPad Prism.

RNA Electrophoretic Mobility Shift

Assay (EMSA) and denaturing urea PAGE

Recombinant full length Dbp5 was purified as above and described previously (Montpetit, Thomsen et al. 2011, Montpetit, Seeliger et al. 2012). RNA binding and EMSA was performed according to previously published assay conditions (Yao, Roser et al. 2007). RNA binding reactions were assembled using binding buffer (20mM HEPES [pH7.4], 100 mM KCl, 10 mM NaCl, 4 mM MgCl₂, 0.2 mM EDTA, 20% glycerol, 1 mM DTT, 0.5% NP-40) in the presence of indicated nucleotide and loaded on to 6% native polyacrylamide el (0.5 x TBE). Gels were stained with Apex gel stain to visualize RNA. Mean integrated intensity of bands were quantified in FIJI (Schindelin, Arganda-Carreras et al. 2012) and normalized to background signal for each respective well. Bound fraction was calculated and fit to a one-site binding equation in GraphPad Prism.

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Reviewer #1 (Public Review):

Summary:

This study focuses on the defining cellular pathways critical for tRNA export from the nucleus. While a number of these pathways have been identified, the observation that the primary transport receptors identified thus far (Los1 and Msn5) are not essential and that cells are viable even when both the genes are deleted supports the idea that there are as yet unidentified mediators of tRNA export from the nucleus. This study implicates the helicase Dbp5 in one of these parallel pathways arguing that Dbp5 works in a pathway that is independent of Los1 and/or Msn5. The authors present genetic data to support this conclusion. At least one result suggests that the idea of these pathways in parallel may be too simplistic as deletion of the *LOS1* gene, which is not essential decreases the interaction of tRNA export substrate with Dbp5 (Figure 2A). If the two pathways were working in parallel, one might have expected removing one pathway to lead to an increase in the use of the other pathway and hence the interaction with a receptor in that pathway. The authors provide solid evidence that Dbp5 interacts with tRNA directly and that the addition of the factor Gle1 together with the previously identified co-factor InsP6 can trigger helicase activity and release of tRNA. The combination of in vivo studies and biochemistry provides evidence to consider how Dbp5 contributes to the export of tRNA and more broadly adds to the conversation about how coding and non-coding RNA export from the nucleus might be coordinated to control cell physiology.

Strengths and weaknesses:

A major strength of this manuscript is the multi-pronged approach to explore a potential role for the helicase Dbp5 in one of the multiple export pathways for tRNA from the nucleus.

The obvious missing experiment here with respect to genetics is the test of whether deletion of the *MSN5* gene in the cells, which combines deletion of *LOS1* and the *dbp5_R423A* allele, shown in Figure 1D would be lethal. This key experiment would lend substance to the argument that Dbp5 functions in a tRNA export pathway that is parallel to the Los1 and Msn5 pathways.

While some of the binding assays show rather modest band shifts (Figure 4B for example), the data in Figure 4A showing that there is no binding detected unless a non-hydrolyzable ATP analogue is employed, argues for specificity in nucleic acid binding. The question that does arise is whether the binding is specific for tRNA.

With the exception of the binding studies, which also employ a mixture of yeast tRNAs, this study relies primarily on a single tRNA species to come to the conclusions drawn. Many other studies have used multiple tRNAs to explore whether pathways characterized are generalizable to other tRNAs.

The authors provide evidence of a model where the helicase Dbp5 plays a role in tRNA export from the nucleus. Further evidence is required to determine whether Dbp5 could function in the same pathway as the previously defined tRNA export receptors, Los1 and Msn5. There are genetic tests that could be performed to explore this question. Some of the biochemistry presented would show when Los1 is absent that the interaction of Dbp5 with tRNA decreases, which could support a model where Dbp5 plays a role in coordination with Los1.

This work allows insight into key questions which still remain about the multiple pathways that are required for tRNA trafficking as well as how transport pathways for coding and non-coding RNAs might be coordinated. These questions are important as many of these pathways may be regulated in response to cellular conditions or during development and defining the fundamental pathways will be critical to understanding these dynamic processes.

Reviewer #2 (Public Review):

This submission is about the role of Dbp5/Gle1 in tRNA export. The manuscript provides data showing that Dbp5/Gle1 are involved in tRNA export from the nucleus which is an essential process critical to translation. The authors provide data that largely supports conclusions, however, there are some pieces of data that are misinterpreted. (Figure 1A and B look the same; in Fig 1E, the DAPI staining is abnormal; in Fig 4 the bands can't be seen.)

Additionally, the methods used are fairly standard so the article does not contain any new technical achievements.

Author Response

We would like to thank reviewers and editors for their thoughtful and constructive review of our manuscript. Below we have provided responses to specific points in the reviewers' comments and eLIFE assessment, highlighting areas of the manuscript that will be edited for clarity and where efforts will be made to provide data to address reviewer concerns upon a future resubmission.

eLife assesment:

The authors report that Dbp5 functions in parallel with Los1 in tRNA export, in a manner dependent on Gle1 and requiring the ATPase cycle of Dbp5, but independent of Mex67, Dbp5's partner in mRNA export. The evidence for this conclusion is still incomplete, as is the biochemical evidence that Dbp5 interacts directly with tRNA in vitro with Gle1 and co-factor InsP6 triggering Dbp5 ATPase activity in the Dbp5-tRNA complex. The evidence that Dbp5 interacts with tRNA in cells independently of Los1, Msn5 and Mex67 is, however, solid.

We intend to edit the text to make clear our conclusions and accommodate clarifications on a few details of this assessment.

(1) We would clarify that our data supports a model in which Dbp5 recruitment to tRNA is independent of Mex67 as an adapter in cells; however, this does not mean that Mex67 and Dbp5 do not still co-function in tRNA export. For example, it is possible Dbp5 and Mex67 could still co-function in the same pathway, but instead of Dbp5 working down stream of Mex67, Dbp5 may in fact work upstream as an adapter for Mex67. Edits to the text will be made to ensure this distinction is clear and highlight the possibility for future investigation to elucidate this relationship.

(2) We would like to highlight that based on structural and biochemical data detailing synergistic activation of Dbp5 ATPase cycle by Gle1/InsP6 and single stranded RNA, it is difficult to imagine a scenario where the apparent synergistic activation of Dbp5 ATPase cycle by tRNA and Gle1/InsP6 (Figure 5) is achieved independent of direct RNA binding. For this reason, we still support the claim that the observed synergistic activation, in combination with other in-vivo and in-vitro data provided in the manuscript, support a model where Dbp5 directly binds tRNA. However, we intend to edit the text to highlight this nuance and potential alternative conclusions based on reviewer feedback.

Reviewer #1 (Public Review):

"At least one result suggests that the idea of these pathways in parallel may be too simplistic as deletion of the LOS1 gene, which is not essential decreases the interaction of tRNA export substrate with Dbp5 (Figure 2A). If the two pathways were working in parallel, one might have expected removing one pathway to lead to an increase in the use of the other pathway and hence the interaction with a receptor in that pathway.... The obvious missing experiment here with respect to genetics is the test of whether deletion of the MSN5 gene in the cells, which combines deletion of LOS1 and the dbp5_R423A allele, shown in Figure 1D would be lethal.... The authors provide evidence of a model where the helicase Dbp5 plays a role in tRNA export from the nucleus. Further evidence is required to determine whether Dbp5 could function in the same pathway as the previously defined tRNA export receptors, Los1 and Msn5. There are genetic tests that could be performed to explore this question. Some of the biochemistry presented would show when Los1 is absent that the interaction of Dbp5 with tRNA decreases, which could support a model where Dbp5 plays a role in coordination with Los1"

We agree that this is an important point that should be made clear and discussed in the text. We also agree that further experiments would be needed to be to confirm Dbp5 functions broadly in tRNA export in parallel to both Msn5 and Los1. We will aim to address these points in resubmission and discuss possible alternative conclusions of the presented results.

Reviewer #1 (Public Review):

"While some of the binding assays show rather modest band shifts (Figure 4B for example), the data in Figure 4A showing that there is no binding detected unless a non-hydrolyzable ATP analogue is employed, argues for specificity in nucleic acid binding. The question that does arise is whether the binding is specific for tRNA."

The specificity of the in-vitro interactions of Dbp5 are an important point of discussion. We will work to expand the topic of specificity of the in-vitro experiments during resubmission.

Reviewer #1 (Public Review):

"With the exception of the binding studies, which also employ a mixture of yeast tRNAs, this study relies primarily on a single tRNA species to come to the conclusions drawn. Many other studies have used multiple tRNAs to explore whether pathways characterized are generalizable to other tRNAs."

It was previously shown that Dbp5 functions to support the export of multiple tRNA species (<https://doi.org/10.7554/eLife.48410>). As such, we agree that additional tRNAs should be tested to explore whether phenotypes reported here are also generalizable to other tRNAs. We will add data targeting additional tRNAs during resubmission.

Reviewer #2 (Public Review):

"there are some pieces of data that are misinterpreted. (Figure 1A and B look the same; in Fig 1E, the DAPI staining is abnormal; in Fig 4 the bands can't be seen.)"

Figure 1A and B represent separate experiments, showing that deletion of Los1 does not alter Dbp5 localization and conversely loss of Dbp5 does not alter Los1 localization. As such localization patterns under loss-of-function conditions look the same as wild-type localization for each protein respectively as noted. We believe that we have come to the same conclusion as the reviewer on Figure 1A and B (and this data is not misinterpreted), but also understand

this panel will need to be adjusted for clarity and readability. We will make efforts to edit this figure and accompanying text make the data and conclusions clearer, including addressing the EMSAs in figure 4 and associated text for clarity.